

Aravinda Boovaraghavan

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Education

Carnegie Mellon University

Master of Information Systems Management

Pittsburgh, PA

August 2024 - December 2025

Coursework: ML in Production, Intro to ML, Gen AI, Statistics, Agile Methods, Product Management, Digital Transformation

Vellore Institute of Technology

Bachelor of Technology - Computer Science and Engineering

Chennai, India

June 2019 - July 2023

Skills

Programming & ML: Python, Java, SQL, PyTorch, TensorFlow, scikit-learn, Pandas, NumPy

Data: REST APIs, SDK Integration, Flask, PostgreSQL, MongoDB, API Debugging, Multi-source Data Integration, Entity Resolution

AI: LLMs (Claude, GPT), RAG, Agentic Workflows, Document Processing, Multi-Agent Architectures, Embeddings, Vector DBs, LangChain, LangGraph, LangSmith, Prompt Engineering, Model Evaluation, Claude Code

Cloud & DevOps: AWS, Azure, Git, GitHub, CI/CD, Docker, Kafka, Kubernetes, Grafana

Business & Delivery: Stakeholder Collaboration, Technical Documentation, Agile, SDLC

Research Experience

Carnegie Mellon University

Research Assistant - Professor Mohsen Foroughifar

Pittsburgh, PA

January 2026 - June 2026

- Analyzed 9M+ digital advertising auction records using Python and large-scale data pipelines to study bidding behavior, response time, and competition effects on revenue outcomes and advertiser performance.

Research Assistant - Professor Rahul Telang

May 2025 - October 2025

- Analyzed 1M+ U.S. Copyright Office registrations (1978–2021) using Python and data pipelines to study how major studios adjusted filings across piracy and streaming eras. Built title-matching pipelines achieving 97% match accuracy.

Experience

MCFA Global

AI Systems Management Intern

Philadelphia, PA

June 2026 - Present

- Architected a 3-layer AI pipeline for a federal consulting firm, automating ingestion from 5+ government APIs, applying LLM-based opportunity scoring, and surfacing contract leads 3–18 months before public release, replacing a fully manual process.
- Debugged and re-architected a Python pipeline integrating SAM.gov and USASpending.gov REST APIs under real production constraints, designing a dual-call architecture to resolve API type segregation issues, handling pagination across 1,000+ record result sets, and filtering 40+ NAICS codes across 8 federal agencies independently to deliver targeted contract intelligence daily.
- Used Claude Code and the Claude API as primary development tools to build and iterate the pipeline solo, including migrating the reasoning layer from Gemini to Claude Sonnet 5 without interrupting production use.

A.T Kearney

Student Technology Consultant - Capstone

Pittsburgh, PA

August 2025 - December 2025

- Translated complex multi-source economic data into a Streamlit decision-support tool and clear investment recommendations for a private equity client, converting technical time-series findings into language a non-technical stakeholder team could act on.

PricewaterhouseCoopers (PwC) India

Associate Technology Consultant - Emerging Technologies

Chennai, India

July 2023 - April 2024

- Built and deployed an AI-powered enterprise document intelligence platform using Flask, ReactJS, GPT-3, Llama, and Azure, enabling natural-language querying over internal consulting documents and improving information retrieval for business users.
- Worked with client stakeholders to translate business requirements into scalable analytics dashboards, integrating backend APIs and data workflows to support decision-making across public, official, and administrative users.

Technology Consultant Intern - Emerging Technologies

January 2023 - July 2023

- Performed comprehensive IT due diligence for a healthcare app, identifying 18 critical improvement areas and formulating a 5-milestone technology roadmap to strengthen infrastructure reliability, security, and app performance.

Projects

AI Multimodal Product Intelligence System [Multi-Agent, RAG]

October 2025 - November 2025

- Built an orchestrated 6-agent pipeline using GPT-4o and LangChain to extract structured attributes from unstructured product data via RAG with FAISS retrieval, demonstrating end-to-end document intelligence across ingestion, analysis, and synthesis.
- Designed agentic workflows spanning document analysis, review synthesis, and multi-modal output generation using DALL-E 3 and Gemini Flash 2.5, combining multiple LLM and API integrations in a single production pipeline.

AI-Powered Movie Recommendation System [Machine Learning, MLOps]

August 2025 - December 2025

- Developed a large-scale movie recommendation system (1M users, 27K movies) by training and evaluating matrix factorization models (SVD, ALS) on 2.5M user movie interactions derived from Kafka logs.
- Engineered a fault-tolerant production framework with Flask, Docker, and CI/CD, drift monitoring, and versioning; validated reliability through load testing across 2,000 requests at 100% success and 115.94ms average latency.

Discord Channel Recommendation System [Multi-Agent, LLM Evaluation]

October 2025 - December 2025

- Built a multi-agent recommendation system (LangChain, LangGraph) and a LangSmith evaluation framework, reducing token usage by 36% and latency by 26% while maintaining accuracy.